

Md. Tusher Mollah



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Curriculum Vitae

About Myself

I completed my PhD in Manufacturing Engineering, followed by a Postdoc at the Technical University of Denmark (DTU). My PhD research focused on Computational Fluid Dynamics (CFD) modelling and experimental analysis of Material Extrusion Additive Manufacturing (MEX-AM), including improving layer stability and dimensional accuracy, and integrating reinforcement bars in 3D concrete printing. During my postdoctoral research, I contributed to the OptiCore platform, which aimed to develop cost-effective manufacturing solutions for lightweight wind turbine blades. I developed a CFD model to simulate the vacuum-assisted resin infusion process in sandwich composite structures used in blade manufacturing. In addition, I worked on 3D printing of reinforced composites, geopolymers, and lunar regolith. Overall, I bring a strong background in CFD modelling, additive and composite manufacturing processes, and process optimization. I am currently working as an Assistant Professor at the Islamic University of Technology. I have authored over 40 peer-reviewed publications, with more than 835 citations.

Academic Background

Jul-19 to June-23 **Ph.D. in Manufacturing Engineering**, Department of Civil and Mechanical Engineering (DTU Construct), Technical University of Denmark (DTU), 2800 Kgs. Lyngby, Denmark.

Medium of instruction: **English**.

Ph.D. Thesis: Computational Fluid Dynamics Modelling and Experimental Analysis of Material Extrusion Additive Manufacturing.

- Investigated layer morphology and its stability during multilayer printing in Robocasting.
- Investigated the integration of reinforcement bars and the presence of air voids in the structure during 3DCP.
- Investigated the printing of corners and its precision during filament-based MEX-AM.

Jan-17 to Apr-18 **M.Sc. in Applied Mathematics**, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Gopalganj-8100, Bangladesh.

Result: CGPA 3.88 (on scale 4.00) with **Distinction**. Position: **2nd**. Danish weighted average: **11.2** in Courses and **12.0** in M.Sc. Thesis. Medium of instruction: **English**.

M.Sc. Thesis: Unsteady MHD Bingham Fluid Flow through a Parallel Plate with Ion-Slip and Hall Current.

- Investigated 2D channel flow of non-Newtonian fluid within the magneto-hydrodynamic field.
- Investigated the channel flow between two Riga plates within the electro-magneto-hydrodynamic field.

Jan-13 to Dec-16 **B.Sc. in Mathematics**, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Gopalganj-8100, Bangladesh.

Result: CGPA 3.88 (on scale 4.00) with **Distinction**. Position: **2nd**. Danish weighted average: **11.4** in Courses and **12.0** in B.Sc. Thesis. Medium of instruction: **English**.

B.Sc. Thesis: Numerical Analysis on Unsteady Heat Transfer of Viscous Compressible Boundary Layer Fluid Flow through a Porous Plate with Induced Magnetic Field for an Irrotational System.

Experiences

Dec-25 to Present **Assistant Professor in Mathematics**, Department of Natural Sciences, Islamic University of Technology (IUT), Dhaka, Bangladesh.

Aug-23 to Dec 25 **Postdoctoral Researcher**, Department of Civil and Mechanical Engineering, Section of Digital Building Technologies, Technical University of Denmark (DTU), 2800 Kgs. Lyngby, Denmark.

Feb-23 to Jul 23 **Research Assistant**, Department of Civil and Mechanical Engineering, Section of Manufacturing Engineering, Technical University of Denmark (DTU), 2800 Kgs. Lyngby, Denmark.

Mar-22 to Jun-22 **Visiting Researcher**, Wolfson School of Mechanical, Electrical and Manufacturing Engineering, Loughborough University, Loughborough LE11 3TU, United Kingdom.

- Jul-19 to Feb-23 **PhD Fellow**, Department of Civil and Mechanical Engineering, Section of Manufacturing Engineering, Technical University of Denmark (DTU), 2800 Kgs. Lyngby, Denmark.
- Jan-19 to Jun-19 **Lecturer**, Department of Mathematics at European University of Bangladesh, 2/4 Gabtoli, Mirpur, Dhaka-1216, Bangladesh.
- Nov-18 to Jun-19 **Adjunct Lecturer in Mathematics**, Department of Computer Science and Engineering at Central University of Science and Technology, Plot-A/5, Block-A, Mirpur-14, Dhaka, Bangladesh.
- Jun-18 to Dec-18 **Research Assistant** at Khulna University, Khulna-9208, Bangladesh; under *Prof. Dr. Md. Mahmud Alam*, Mathematics Discipline, Khulna University, Khulna-9208, Bangladesh.
- Jul-17 to Jun-18 **Research Fellow** at National Science and Technology (NST) under the Ministry of Science and Technology, Government of People's Republic of Bangladesh with *Prof. Dr. Md. Mahmud Alam*, Mathematics Discipline, Khulna University, Khulna-9208, Bangladesh.

Research Areas and Professional Profiles

I have published more than 40 papers with over 835 citations (h-index 15, i10-index 21) within the fields of Material Extrusion Additive Manufacturing, Fiber Orientation, Multi-material Printing, Materials Rheology, Composite Manufacturing, Computational Fluid Dynamics, Porous Media Flow Modelling, Fluid Mechanics and Heat Transfer, and Numerical Analysis. Feel free to check out the professional profiles below:

- Google Scholar: <https://scholar.google.com/citations?user=QFkMT0cAAAAJ&hl=en>
- ResearchGate: https://www.researchgate.net/profile/Md_Tusher_Mollah3
- LinkedIn: <https://www.linkedin.com/in/md-tusher-mollah-bsmrstu/>
- ORCID: <https://orcid.org/0000-0001-8678-8908>

Featured Publications

1. B. Šeta, M. Sandberg, M. Brander, M.T. Mollah, V. Kumar, J. Spangenberg.. "Subvoxel control of fiber orientation via multidirectional shearing in 3D printing." *Advanced Science*, (2025) e11008.
2. M.T. Mollah, M. Larionov, R.S. Pierce, J. Spangenberg. "Computational fluid dynamics modelling of vacuum-assisted resin infusion in composite sandwich panels during wind turbine blade manufacturing." *In 21st European Conference on Composite Materials*, 3 (2024) 281-288.
3. M.T. Mollah, R. Comminal, W.R. Leal da Silva, B. Šeta, J. Spangenberg. "Computational fluid dynamics modelling and experimental analysis of reinforcement bar integration in 3D concrete printing." *Cement and Concrete Research*, 173 (2023) 107263.
4. M.T. Mollah, R. Comminal, M.P. Serdeczny, B. Šeta, J. Spangenberg. "Computational analysis of yield stress buildup and stability of deposited layers in material extrusion additive manufacturing." *Additive Manufacturing*, 71 (2023) 103605.
5. M.T. Mollah, R. Comminal, M.P. Serdeczny, D.B. Pedersen, J. Spangenberg. "Stability and deformations of deposited layers in material extrusion additive manufacturing." *Additive Manufacturing*, 46 (2021) 102193. doi:10.1016/j.addma.2021.102193.

Supervision Experiences

Postgraduate Projects

1. *Fiber orientation in large area 3DCP*, Master's Thesis, DTU, 2800 Kgs. Lyngby, Denmark.
2. *Experiment and Simulation of Wet-on-Solidifying 3D Printing*, Master's Thesis, DTU, 2800 Kgs. Lyngby, Denmark.
3. *Experimental and numerical analysis of big area additive manufacturing*, Master's Thesis, DTU, 2800 Kgs. Lyngby, Denmark.
4. *Numerical Simulation of Material Extrusion Process*, Master's semester project, DTU, 2800 Kgs. Lyngby, Denmark.
5. *Experimental and numerical analysis of 3D bio-printing*, Master's semester project, DTU, 2800 Kgs. Lyngby, Denmark.

Undergraduate Projects

1. *3D printing and strength analysis of concrete with reinforcement*, Bachelor Thesis, DTU, 2800 Kgs. Lyngby, Denmark.
2. *3D printing in construction focusing on overhang structures*, Special course for Erasmus+ student, DTU, 2800 Kgs. Lyngby, Denmark.

Professional Membership

IDA, Danish Center for Applied Mathematics and Mechanics (DCAMM), Bangladesh Mathematical Society, and The International Union of Laboratories and Experts in Construction Materials, Systems and Structures (RILEM, from the name in French)

Award, Fellowship, and Certification

- Best Paper Award Symposium Best Papers "Numerical Predictions of Bottom Layer Stability in Material Extrusion Additive Manufacturing" at Solid Freeform Fabrication 2021: Proceedings of the 32nd Annual International Solid Freeform Fabrication Symposium – An Additive Manufacturing Conference, Texas, USA, Aug 2021.
- International Presentation Received multiple travel grants totaling over **40,000 DKK** from Thomas B. Thriges Fond and Otto Mønstedts Fond between 2022 and 2025 to support conference participation in the USA, France, and Germany.
- Research Funding Awarded **45,000 DKK** (25,000 DKK from Thomas B. Thriges Fond and 20,000 DKK from Otto Mønstedts Fond) to support a research stay at Loughborough University, UK, in 2022.
- Award 6th Undergraduate National Mathematics Olympiad in Khulna Division, Khulna University on November 2014. *Awarded for Superior Performance.*
- Research Fellowship National Science and Technology (NST) Fellowship (2017-2018), Ministry of Science and Technology, Government of People's Republic of Bangladesh.
- Certification BCC Certification of Leveraging ICT for Growth, Employment and Governance. Certification Type: 1 Foundation Skills Training. Skills Type: Soft Skill(60 hours program) & Web Designing(100 hours program). Status: Certificated from NASSCOM.
- Certification 2 Flow-3D POST, Creating Optimal Workflows Training in Munich, Germany. One day training by the commercial CFD software company Flow Science, Inc.
- Certification 3 OpenFOAM training to advance CFD solver customization, case setup automation (via Docker), and visualization workflows in ParaView. One week training Roskilde University, Denmark.

Technical Skills

- Programming MATLAB, FORTRAN, C/C++.
- Simulation Tools Flow-3D, ANSYS Fluent, OpenFoam, COMSOL Multiphysics.
- CAD SolidWorks, Autodesk Fusion.
- Scripting PHP, CSS, HTML.
- Computing Proficient on Microsoft Word, Excel, Power Point, Adobe Photoshop, ImageJ.

Language Skills

• Bengali: Native or Bilingual Proficiency • English: Professional Working Proficiency • Danish: Elementary Proficiency

References

Dr. Jon Spangenberg

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Dr. Andy Gleadall

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Dr. Douglas E. Smith

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Signature **Dr. Md. Tusher Mollah**
February, 2026

Journal Publications

1. M.T. Mollah, M. Larionov, R.S. Pierce, B. Šeta, J. Spangenberg. "Experimental and Numerical Investigation of Fluid Flow Behavior in Infusion Channels and Reinforcement Fabrics." *Submitted to Polymer Composites*.
2. A.B. Kachalov, P.S. Sánchez, M.T. Mollah, J.M. Ezquerro, J. Spangenberg, B. Šeta. "Numerical Analysis of Coaxially 3D Printed Lunar Habitats: Integrating Regolith and PCM for Passive Temperature Control." *Microgravity Sci. Technol.*, 37, 38(2025). doi.org/10.1007/s12217-025-10194-4.
3. M.T. Mollah, B. Šeta, J. Spangenberg. "A Virtual 3D Printing Framework for Off-Earth Construction." *Acta Astronautica*. doi:10.1016/j.actaastro.2026.02.001
4. B. Šeta, M. Sandberg, M. Brander, M.T. Mollah, V. Kumar, J. Spangenberg. "Subvoxel control of fiber orientation via multidirectional shearing in 3D printing." *Advanced Science*, (2025) e11008. doi:10.1002/advs.202511008.
5. B. Šeta, M. Sandberg, M. Brander, M.T. Mollah, D.K. Pokkalla, V. Kumar, J. Spangenberg. "Numerical modeling of fiber orientation in multi-layer, isothermal material-extrusion big area additive manufacturing." *Additive Manufacturing*, 92 (2024) 104396. doi:10.1016/j.addma.2024.104396.
6. E.A. Khalid, A.K. Issam, M.E. Jai, B. Šeta, M.T. Mollah. "3D concrete printing using computational fluid dynamics: Modeling of material extrusion with slip boundaries." *Journal of Manufacturing Processes*, 118 (2024) 448-459. doi:10.1016/j.jmapro.2024.03.042.
7. M.T. Mollah, R. Comminal, W.R. Leal da Silva, B. Šeta, J. Spangenberg. "Computational fluid dynamics modelling and experimental analysis of reinforcement bar integration in 3D concrete printing." *Cement and Concrete Research*, 173 (2023) 107263. doi:10.1016/j.cemconres.2023.107263.
8. B. Šeta, M. Sandberg, M. Brander, M.T. Mollah, D.K. Pokkalla, V. Kumar, J. Spangenberg. "Modeling fiber orientation and strand shape morphology in three-dimensional material extrusion additive manufacturing." *Composites Part B: Engineering*, 266 (2023) 110957. doi:10.1016/j.compositesb.2023.110957.
9. M.T. Mollah, R. Comminal, M.P. Serdeczny, B. Šeta, J. Spangenberg. "Computational analysis of yield stress buildup and stability of deposited layers in material extrusion additive manufacturing." *Additive Manufacturing*, 71 (2023) 103605. doi:10.1016/j.addma.2023.103605.
10. H. Moghadasi, M.T. Mollah, D. Marla, H. Saffari, J. Spangenberg. "Computational Fluid Dynamics Modeling of Top-Down Digital Light Processing Additive Manufacturing." *Polymers*, 15 (2023) 2459. doi:10.3390/polym15112459.
11. E.A. Khalid, A.K. Issam, M.T. Mollah, J. Spangenberg. "Numerical modelling of planned corner deposition in 3D concrete printing." *Archives of Materials Science and Engineering*, 121(2) (2023), 71-79.
12. M.P. Serdeczny, R. Comminal, M.T. Mollah, D.B. Pedersen, J. Spangenberg. "Viscoelastic simulation and optimisation of the polymer flow through the hot-end during filament-based material extrusion additive manufacturing." *Virtual and Physical Prototyping*, 17 (2022) 205–219. doi:10.1080/17452759.2022.2028522.
13. M.T. Mollah, R. Comminal, M.P. Serdeczny, D.B. Pedersen, J. Spangenberg. "Numerical Predictions of Bottom Layer Stability in Material Extrusion Additive Manufacturing." *JOM*, 74 (2022) 1096–1101. doi:10.1007/s11837-021-05035-9.
14. M.A. Sayeed, A. Podder, M.T. Mollah, M. Wahiduzzaman, G. Lorenzini, M.M. Alam. "Unsteady MHD Viscous Nanofluid Flow Containing Gyrotactic Microorganisms through a Cylindrical Outer Region." *Journal of Engineering Thermophysics*, 31 (2022) 522–536. doi:10.1134/S1810232822030134.
15. A. Podder, M.T. Mollah, M. Wahiduzzaman, G. Lorenzini, M.M. Alam. "Transverse Magnetic Field Effect on Extracellular Fluid Flow along with a Semi-Infinite Vertical Rotating Porous Plate." *Saudi Journal of Engineering and Technology*, 7 (2022) 218–230. doi:10.36348/sjet.2022.v07i05.003.
16. J. Spangenberg, W.R. Leal da Silva, R. Comminal, M.T. Mollah, T.J. Andersen, H. Stang. "Numerical simulation of multi-layer 3D concrete printing." *RILEM Technical Letters*, 6 (2021) 119–123. doi:10.21809/rilemtechlett.2021.142.
17. M. Parvine, M.T. Mollah, S. Poddar, M. Alam, G. Lorenzini. "EMHD Nanofluid Flow Along A Porous Riga Plate With Thermal Radiation." *Journal of Mechanical Engineering Research and Developments*, 44 (2021) 140–155.
18. M.T. Mollah, H.K. Rasmussen, S. Poddar, M.M. Islam, M. Parvine, M.M. Alam, G. Lorenzini. "Ion-Slip Effects on Bingham Fluid Flowing Through an Oscillatory Porous Plate with Suction." *Mathematical Modelling of Engineering Problems*, IIETA. 8 (2021) 673–681. doi:10.18280/mmep.080501.
19. M.T. Mollah, S. Poddar, M.M. Islam, M.M. Alam. "Non-isothermal Bingham fluid flow between two horizontal parallel plates with Ion-slip and Hall currents." *SN Applied Sciences*, 3 (2021) 115. doi:10.1007/s42452-020-04012-2.
20. M.T. Mollah, R. Comminal, M.P. Serdeczny, D.B. Pedersen, J. Spangenberg. "Stability and deformations of deposited layers in material extrusion additive manufacturing." *Additive Manufacturing*, 46 (2021) 102193. doi:10.1016/j.addma.2021.102193.
21. S. Khatun, M.M. Islam, M.T. Mollah, S. Poddar, M.M. Alam. "EMHD radiating fluid flow along a vertical Riga plate with suction in a rotating system." *SN Applied Sciences*, 3 (2021) 452. doi:10.1007/s42452-021-04444-4.

22. M.S. Hasan, M.T. Mollah, D. Kumar, R.N. Mondal, G. Lorenzini. "Effects of Rotation on Transient Fluid Flow and Heat Transfer Through a Curved Square Duct: The Case of Negative Rotation." *International Journal of Applied Mechanics and Engineering*, 26 (2021) 29–50. doi:10.2478/ijame-2021-0048.
23. M.P. Serdeczny, R. Comminal, M.T. Mollah, D.B. Pedersen, J. Spangenberg. "Numerical modeling of the polymer flow through the hot-end in filament-based material extrusion additive manufacturing." *Additive Manufacturing*, 36 (2020) 101454. doi:10.1016/j.addma.2020.101454.
24. M.T. Mollah. "EMHD Laminar Flow of Bingham Fluid Between Two Parallel Riga Plates." *International Journal of Heat and Technology*, IETA. 37 (2019) 641–648. doi:10.18280/ijht.370236.
25. M.T. Mollah, M.M. Islam, S. Khatun, M. Ferdows, M.M. Alam. "MHD Generalized Couette Flow and Heat Transfer on Bingham Fluid Through Porous Parallel Plates." *Mathematical Modelling of Engineering Problems*, IETA. 6 (2019) 483–490. doi:10.18280/mmep.060402.
26. S. Khatun, M.T. Mollah, Mst.S. Akter, M. Islam, M.M. Alam. "EMHD Couette Flow of Bingham Fluid Through a Porous Parallel Riga Plates with Thermal Radiation." *Modelling, Measurement and Control B*, 88 (2019) 64–71. doi:10.18280/mmcb.882 – 409.
27. M.M. Islam, M.T. Mollah, S. Khatun, M. Ferdows, M.M. Alam. "Unsteady Viscous Incompressible Bingham Fluid Flow through a Parallel Plate." *Inventions*, 4 (2019) 51. doi:10.3390/inventions4030051.
28. M.T. Mollah, M.M. Islam, M.M. Alam. "Hall and ion-slip effects on unsteady MHD Bingham fluid flow with suction." *Modelling, Measurement and Control B*, 87 (2018) 221–229. doi:10.18280/mmcb.871 – 402.
29. M.M. Islam, M.T. Mollah, M.S. Hasan, M.M. Alam. "Numerical Solution of Unsteady Viscous Compressible Fluid Flow along a Porous Plate with Induced Magnetic Field." *Modelling, Measurement and Control B*, 86 (2017) 850–863. doi.org/10.18280/mmcb.860403.

Monograph

1. M.T. Mollah (2023) "Computational Fluid Dynamics Modelling and Experimental Analysis of Material Extrusion Additive Manufacturing." *Ph.D.-Thesis*, Technical University of Denmark.
2. M.T. Mollah (2018) "Unsteady MHD Bingham Fluid Flow through a Parallel Plate with Ion-Slip and Hall Current." *M.Sc.-Thesis*, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Bangladesh.
3. M.T. Mollah (2016) "Numerical Analysis on Unsteady Heat Transfer of Viscous Compressible Boundary Layer Fluid Flow through a Porous Plate with Induced Magnetic Field for an Irrotational System." *B.Sc.- Project Thesis*, Bangabandhu Sheikh Mujibur Rahman Science and Technology University, Bangladesh.

Conference Publications

1. M.T. Mollah, J. Wonterghem, H. Rasmussen, D. Meng, B. Šeta, S. Rahemipoor, D. Meile, N. Ranjbar, J. Spangenberg. "Computational Fluid Dynamics Modeling and Experimental Analysis of Rapid Curing in 3D Printing for Structural Applications." In *Solid Freeform Fabrication 2025: Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium*, p840-851.
2. D. Meile, C.E. Hansen, Y. Jacquet, M.T. Mollah, B. Šeta, J. Spangenberg. "Computational Fluid Dynamics Modeling of Sequential Multi-Material Extrusion Additive Manufacturing ." In *Solid Freeform Fabrication 2025: Proceedings of the 36th Annual International Solid Freeform Fabrication Symposium*, p852-863.
3. M.T. Mollah, M. Larionov, R.S. Pierce, J. Spangenberg. "Computational fluid dynamics modelling of vacuum-assisted resin infusion in composite sandwich panels during wind turbine blade manufacturing." In *21st European Conference on Composite Materials*, 3 (2024) 281-288. doi:10.60691/yj56-np80
4. M. Larionov, M.T. Mollah, J. Spangenberg R.S. Pierce. "Optimization of core groove geometry for the manufacture and operation of composite sandwich structures in wind turbine blades." In *21st European Conference on Composite Materials*, 5 (2024) 278-285. doi:10.60691/yj56-np80
5. N. Haghighat, M.T. Mollah, S.Y. Sannerud, J. Boyer, J. Spangenberg, S. Jacobsen. "Rheology and printability of cement paste modified with filler from manufactured sand." In *4th RILEM International Conference on Concrete and Digital Fabrication, Technische Universität Braunschweig*, (2024).
6. M.T. Mollah, A. Moetazedian, A. Gleadall, J. Yan, W.E. Alphonso, R. Comminal, B. Šeta, T. Lock, J. Spangenberg. "Investigation on corner precision at different corner angles in material extrusion additive manufacturing: An experimental and computational fluid dynamics analysis." In *Solid Freeform Fabrication Symposium 2022: 33rd Annual Meeting*, The University of Texas at Austin, 2022, pp. 872–881.
7. B. Šeta, M. Sandberg, M. Brander, M.T. Mollah, D.K. Pokkalla, V. Kumar, J. Spangenberg. "Numerical modeling of fiber orientation in additively manufactured composites." In *IOP Conference Series: Materials Science and Engineering*, vol. 1293, no. 1, p. 012033. IOP Publishing, 2023.

8. M.T. Mollah, M.P. Serdeczny, R. Comminal, B. Šeta, M. Brander, J. Spangenberg. "A Numerical Investigation of the Inter-Layer Bond and Surface Roughness during the Yield Stress Buildup in Wet-On-Wet Material Extrusion Additive Manufacturing." In *2022 Summer Topical Meeting: Advancing Precision in Additive Manufacturing*, American Society for Precision Engineering, 2022, pp. 131–134.
9. R. Tonello, M.T. Mollah, A. Strandlie, J. Spangenberg, D.B. Pedersen, J.R. Frisvad. "Influence of the Printing Direction on the Surface Appearance in Multi-Material Fused Filament Fabrication." Accepted in *A Lecture Notes on Computer Science book*.
10. E.A. Khalid, A.K. Issam, M.T. Mollah, J. Spangenberg. "Computational Fluid Dynamics Modelling of Mortar Extrusion and Path Planning Strategy at the Corner for 3D Concrete Printing." In *Casablanca International Conference on Additive Manufacturing*, November 23-24, 2022, Casablanca, Morocco.
11. B. Šeta, M. Brander, M.T. Mollah, V. Kumar, D.K. Pokkalla, S. Kim, J. Spangenberg. "Modelling of Additive Manufacturing–Compression Molding Process using Computational Fluid Dynamics." In *2022 Summer Topical Meeting: Advancing Precision in Additive Manufacturing*, American Society for Precision Engineering, TN, USA, 2022, pp. 119–123.
12. J. Spangenberg, W.R. Leal Da Silva, M.T. Mollah, R. Comminal, T. Juul Andersen, H. Stang. "Integrating Reinforcement with 3D Concrete Printing: Experiments and Numerical Modelling." In *Third RILEM International Conference on Concrete and Digital Fabrication*, R. Buswell, A. Blanco, S. Cavalaro, P. Kinnell (Eds.), Springer International Publishing, Cham, 2022, pp. 379–384. doi:10.1007/978-3-031-06116-5_56.
13. B. Šeta, M.T. Mollah, V. Kumar, D.K. Pokkalla, S. Kim, A.A. Hassen, J. Spangenberg. "Modelling Fiber Orientation during Additive Manufacturing–Compression Molding Processes." In *Proceedings of the 33rd Annual International Solid Freeform Fabrication Symposium*, The University of Texas at Austin, 2022, pp. 906–919.
14. M.T. Mollah, M.M. Islam, M. Ferdows, M.M. Alam. "Bingham fluid flow through oscillatory porous plate with ion-slip and hall current." In *AIP Conference Proceedings*, AIP Publishing, 2121(1), 2021, p. 050011. doi:10.1063/1.5115898.
15. M.M. Islam, S. Khatun, M.T. Mollah, M.M. Alam. "Fluid flow along the Riga plate with the influence of magnetic force in a rotating system." In *AIP Conference Proceedings*, AIP Publishing, 2121(1), 2021, p. 050002. doi:10.1063/1.5115889.
16. Mst.S. Akter, M.R. Islam, M.T. Mollah, M.M. Alam. "Hall effects on Casson fluid flow along a vertical plate." In *AIP Conference Proceedings*, AIP Publishing, 2121(1), 2021, p. 040004. doi:10.1063/1.5115875.

Conference Presentations

1. "Computational Fluid Dynamics Modeling and Experimental Analysis of Rapid Curing in 3D Printing for Structural Applications.", paper presented in *2025 International Solid Freeform Fabrication Symposium*, University of Texas at Austin, USA, Aug 2025.
2. "Computational fluid dynamics modelling of vacuum-assisted resin infusion in composite sandwich panels during wind turbine blade manufacturing", paper presented in *ECCM21 – 21st European Conference on Composite Materials 2024*, Nantes, France, Jul 2024.
3. "Strategies for Geometrically Stable Printing in Material Extrusion Additive Manufacturing", abstract presented in *FLOW-3D World Users Conference 2024*, Hamburg, Germany, Jun 2024.
4. "Computational Fluid Dynamics Modelling and Experimental Analysis of Resin Infusion Performance in Composite Sandwich Panels for Wind Turbine Blade Production", poster presented in *FLOW-3D World Users Conference 2024*, Hamburg, Germany, Jun 2024.
5. "Computational Fluid Dynamics Modelling of Material Extrusion Additive Manufacturing", poster presented in *DTU Construct Department Seminar 2022*, Helsingør, Denmark, Sep 2022.
6. "Investigation on corner precision at different corner angles in material extrusion additive manufacturing: An experimental and computational fluid dynamics analysis", paper presented in *2022 International Solid Freeform Fabrication Symposium*, University of Texas at Austin, USA, Jul 2022.
7. "Computational fluid dynamics modelling of the dimensional stability of deposited layers in material extrusion additive manufacturing", abstract presented *FLOW-3D World Users Conference 2022*, Munich, Germany, May 2022.
8. "Numerical Predictions of Bottom Layer Stability in Material Extrusion Additive Manufacturing", paper presented *2021 International Solid Freeform Fabrication Symposium*, University of Texas at Austin, USA, Aug 2021.
9. "Bingham Fluid Flow through Oscillatory Porous Plate with Ion-Slip and Hall Current", paper presented in *8th BSME International Conference on Thermal Engineering 2018*, BUET, Dhaka, Bangladesh.